

A dual-functional additive for crystallization regulated and defect passivated stable inverted perovskite solar cells

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ABSTRACT

The main problems holding back inverted perovskite solar cells (PSCs) are poor crystal formation and too many defects inside the perovskite layer. We address both issues by introducing 4-aminobutylphosphonic acid, which effectively releases residual tensile stress and passivates film defects, leading to enhanced efficiency and stability. We report 4-aminobutylphosphonic acid as a multifunctional additive where the phosphonic acid ($-\text{PO}_3\text{H}_2$) and amine ($-\text{NH}_2$) groups coordinatively passivate undercoordinated Pb^{2+} and I^- vacancies, as supported by GIXRD measurements showing 64% reduction in residual tensile stress, and time-resolved photoluminescence shows enhanced carrier lifetime. The optimized devices achieve a champion power conversion efficiency (PCE) of 24.1% and retain 85% of its initial PCEs after 840 h under 1-sun illumination at 65 °C. This molecular engineering approach provides a facile route to concurrently boost efficiency and stability in inverted PSCs.

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Inverted perovskite solar cells (PSCs) have emerged as a promising architecture for next-generation photovoltaics due to their compatibility with tandem solar cells,^{1,2} low-temperature processing,³ and remarkable power conversion efficiencies approaching those of crystalline silicon.⁴ Even with these changes, defects in the perovskite (PVK) layer still affect the performance and stability of PSCs, especially at grain boundaries (GBs) and interfaces.⁵ These flaws act as centers for non-radiative recombination, and they happen when lead ions and halide vacancies cannot work together. This procedure impedes charge mobility, thereby accelerating the breakdown process of the device.⁶ Beyond defect mitigation, residual stress within PVK films induced during thermal annealing due to mismatched thermal expansion and volatile organic cation loss has emerged as a significant source of instability in PSCs.^{7,8} The management of this residual strain through interface and bulk engineering is therefore a critical frontier for enhancing device longevity and performance.⁹ This mechanical stress accelerates phase degradation and directly impacts charge carrier mobility, making its alleviation crucial for high-performance devices.^{10,11}

Among the major strategies for improving PSC performance, additive engineering has proven to be an effective strategy to mitigate these issues by modulating the crystallization process and passivating electronic trap states.^{12–14} Such strategies are essential to overcome the persistent challenges that currently impede the commercialization pathway of PVK photovoltaics.¹⁵ Recent studies have demonstrated that multi-functional additives can simultaneously suppress defect states and alleviate residual stress, leading to more stable PVK films with improved optoelectronic properties.^{16,17} Multifunctional additives with diverse functional groups can simultaneously target multiple defect types, thereby improving film quality and device durability.^{18,19} Various organic molecules, including small molecules,²⁰ polymers,²¹ and multifunctional organic salts,²² have been introduced into the PVK precursor solution to tailor film morphology and reduce defect density. In particular, molecules with multiple functional groups offer enhanced passivation due to their strong coordination ability with undercoordinated Pb^{2+} and capacity to modulate film stress.^{16,23}

The pursuit of high-performance PSCs has revealed two particularly effective chemical strategies for defect mitigation. The first relies on phosphonic acid groups, which can strongly anchor to undercoordinated Pb^{2+} ions, as demonstrated in passivators such as 3-phosphonopropanoic acid.²⁴ The second strategy involves amine groups, exemplified by additives like ethylenediamine,²⁵ which interact with halide-vacancy sites and influence crystallization kinetics. Building on the complementary roles of these two functional groups, we designed a bifunctional molecule, 4-aminobutylphosphonic acid, that integrates both effects within a single passivator. While 4-aminobutylphosphonic acid (4-ABPA) has been recently reported as a surface passivator for mixed Pb-Sn PVK,²⁶ our work demonstrates its distinct role as a bulk additive in inverted PSCs, where it simultaneously modulates crystallization kinetics, suppresses bulk defects, and reduces residual tensile stress key factors governing device performance and stability. The phosphonic acid terminus strongly coordinates with undercoordinated Pb^{2+} ions, passivating cationic defects, while the amine terminus interacts with anionic defect sites associated with I^- vacancies. This dual-action approach enables simultaneous mitigation of the primary defect types, effectively suppressing non-radiative recombination pathways.

When incorporated directly into the PVK precursor, 4-ABPA not only passivates defects but also alleviates residual lattice strain, facilitating the formation of smoother films with larger grains and improved structural uniformity. Devices incorporating 4-ABPA demonstrate enhanced performance, achieving a power conversion efficiency (PCE) of 24.1% and retaining over 85% of their initial efficiency after 840 h of continuous operation. These results indicate that 4-ABPA serves as an

effective and versatile passivation molecule, providing a promising route to improving both efficiency and operational stability in inverted PSCs. These findings highlight the broader potential of molecularly engineered bifunctional passivators as a practical strategy for enhancing the durability and performance of PVK photovoltaics.

In this study, 4-ABPA was incorporated into the precursor solution of the mixed-cation PVK $\text{Cs}_{0.05}\text{FA}_{0.75}\text{MA}_{0.2}\text{PbI}_3$ as a dual-functional additive to simultaneously passivate defects and regulate crystallization. The multifunctionality of 4-ABPA arises from its molecular structure: a $-\text{PO}_3\text{H}_2$ group that forms coordinative bonds with undercoordinated Pb^{2+} ions and a terminal $-\text{NH}_2$ group capable of hydrogen bonding with organic cations (FA^+ , MA^+) and halide anions (I^-). The $-\text{PO}_3\text{H}_2$ group ionizes to produce $\text{P}-\text{O}^-$, which strongly interacts with unsaturated Pb^{2+} sites, significantly reducing trap densities, as reported previously.^{27,28} Similar to phosphonic acid-based modifiers,²⁴ 4-ABPA interacts with Pb^{2+} through Lewis acid-base interactions and stabilizes the PVK lattice via hydrogen bonding. These synergistic interactions enable 4-ABPA to control nucleation, crystal growth, and defect passivation at grain boundaries, thereby decreasing non-radiative recombination and improving charge transport. Figure 1(a) illustrates how 4-ABPA interacts with the three-dimensional PVK framework. Fourier transform infrared (FTIR) spectroscopy further substantiates the emergence of chemical interactions subsequent to the introduction of 4-ABPA to PbI_2 . The N-H stretching peaks shift from 1662.55 to 1602.76 cm^{-1} , and the P=O groups shift significantly from 1238.23 to 1234.37 cm^{-1} , 1168.80 to 1163.01 cm^{-1} , and 1130.22 to 1128.30 cm^{-1} , as shown in Fig. 1(b). The P-O stretching peaks also exhibit noticeable shifts, indicating

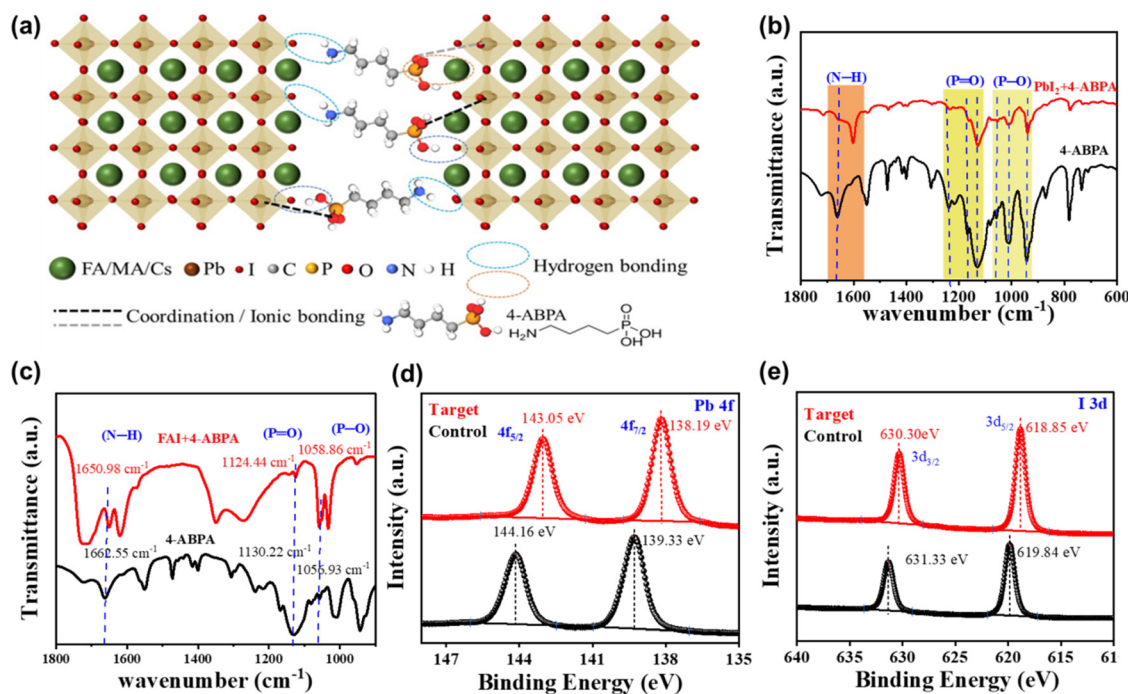


FIG. 1. Structural characterization and molecular interactions: comparison between control and 4-ABPA-modified PVK films. (a) Schematic illustration of the molecular interaction between 4-ABPA and the PVK lattice. (b) FTIR spectra of 4-ABPA and 4-ABPA + PbI_2 . (c) FTIR spectra of 4-ABPA and 4-ABPA + FAI. (d) XPS spectra of Pb 4f, and (e) XPS spectra of I 3d for control and 4-ABPA-modified PVK films.

multiple bonding pathways for the phosphonic groups with the PbI_6^{4-} octahedron, suggesting the formation of strong bonds.^{24,26} The enlarged spectra display enhanced interaction signals (refer to Fig. S1 in the [supplementary material](#)). Additionally, the interaction of 4-ABPA with formamidinium iodide (FAI) results in the N–H and C–N stretching peaks shifting to lower and higher wavenumbers, respectively. This indicates that hydrogen bonding occurs between 4-ABPA and the organic cations [Fig. 1(c)].²⁹

To further investigate the chemical interactions between PVK and with and without 4-ABPA (marked target and control in the figures) components, x-ray photoelectron spectroscopy (XPS) was performed to analyze the chemical states of the primary elements within the PVK film. High-resolution peak-deconvoluted Pb 4f spectra [Fig. 1(d)] exhibit a distinct shift in Pb^{2+} doublet toward lower binding energies (Pb 4f_{5/2}: from 144.16 to 143.05 eV, and Pb 4f_{7/2}: from 139.33 to 138.19 eV). This shift indicates an increase in electron density around the Pb^{2+} ions, attributable to the donation of electrons from the N–H, P=O, and P–O functional groups of 4-ABPA.^{24,30} Additionally, the I 3d peaks also shift to lower binding energies (I 3d_{3/2}: from 619.84 to 618.85 eV, and I 3d_{5/2}: from 631.33 to 630.30 eV). This observation suggests the formation of hydrogen bonds between the –OH groups of 4-ABPA and the I^- ions within the PbI_6^{4-} octahedral framework [Fig. 1(e)].^{24,31} These findings confirm the presence of strong coordination and hydrogen bonding interactions, which play a crucial role in effective defect passivation within the PVK layer.

X-ray diffraction (XRD) analysis validated that the addition of 4-ABPA did not modify the crystal structure, as demonstrated by the identical diffraction peaks in the modified films [Figs. 2(a) and S2 for different additive concentrations]. The (001)/(011) crystal planes in the treated films, on the other hand, had a higher intensity ratio than those in the control films. This indicates that the crystals grew larger

primarily in the [001] direction. This orientation enhances charge transport and device performance due to the superior photoelectric properties of the (001) plane and the absence of surface trap states.^{32,33} The full width at half maximum (FWHM) also went down from 0.111° for the control film to 0.107° for the 4-ABPA-treated film (Table S1). Figure S3 in the [supplementary material](#) shows a close-up of the XRD peak that goes along with this change. This suggests that the crystal quality has improved and the grain size has gotten bigger. The small changes in peak intensity ratios provided more evidence that 4-ABPA affects how crystals form. These findings indicate that 4-ABPA improves (001) orientation, lowers defect density, and makes PSC more stable and efficient.

To assess the residual stress in PVK films, depth-resolved grazing incidence XRD (GIXRD) measurements were performed. In the detailed view of the GIXRD profiles [Figs. 2(b) and 2(c)], in control films, increasing the grazing incidence angle (ψ) from 0° to 50° results in a measurable shift toward lower 2θ angles, indicating significant residual tensile stress.³⁴ In contrast, 4-ABPA-modified films showed negligible diffraction peak shifts, indicating effective stress mitigation. Quantitative analysis of d-spacing vs ψ (based on Bragg's law) revealed a substantial slope reduction from 0.00153 (control) to 0.00055 (modified films) [Fig. 2(d)], confirming residual tensile stress relaxation. This corresponds to approximately 64% reduction in residual tensile stress, calculated as $(1 - 0.00055/0.00153) \times 100\%$. The slight shift from left to right may result from a redistribution of tensile and compressive components, suggesting a balanced reduction of residual stress rather than a simple uniaxial relaxation. This effect is attributed to 4-ABPA integration at grain boundaries, where its terminal functional groups stabilize the PVK octahedral framework while flexible alkyl chains accommodate stress through carbon–carbon bond rotation.³⁵ Stress relief observed by GIXRD is attributed to preferential 4-ABPA incorporation at grain boundaries and surfaces, where its terminal

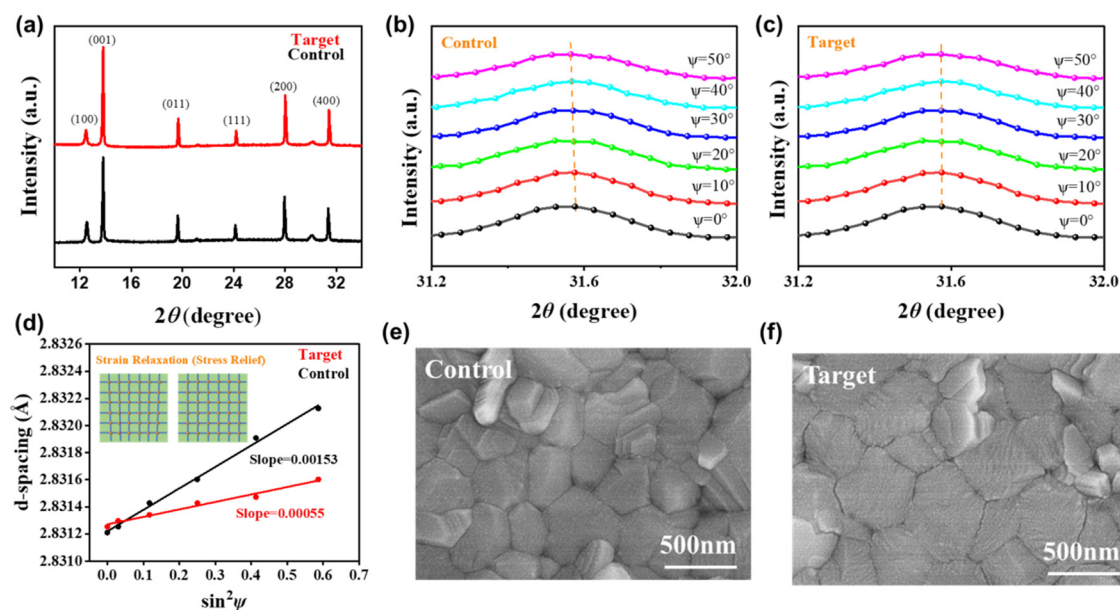


FIG. 2. Structural and morphological characterization of control and 4-ABPA-modified PVK films. (a) XRD patterns. (b) and (c) GIXRD spectra. (d) Residual strain vs $\sin^2\psi$ derived from diffraction peak positions. (e) and (f) Top-view SEM images showing surface morphology of control and 4-ABPA-modified films.

groups stabilize the lattice and the flexible butyl chain accommodates local strain, enabling localized defect passivation.³⁶ Previous studies have shown that the alkyl chain length of passivating amine molecules correlates with defect passivation efficacy in PSCs, where varying the chain length affects surface and grain boundary defect mitigation.³⁷ Longer alkyl chains have been reported to achieve greater surface defect reduction and improved operational stability compared to shorter chains, albeit at the cost of increased insulating character.³⁸ The alleviation of residual stress reduces bulk defect density, improves film quality, and enhances PVK device stability and optoelectronic performance.

The morphological evolution of PVK films upon the incorporation of 4-ABPA was investigated using scanning electron microscopy (SEM). Figures 2(e) and 2(f) show that both the control and modified films have polycrystalline properties. The film treated with 4-ABPA demonstrates a significant increase in grain size compared to the pristine film. The observed grain enlargement can be rationalized by the dual kinetic regulation imparted by 4-ABPA during film crystallization. The strong chelation of the phosphonic acid moiety ($-\text{PO}_3\text{H}_2$) with undercoordinated Pb^{2+} species in the precursor solution is posited to suppress nucleation density, thereby reducing the number of nucleation sites and permitting the growth of larger, more-defined grains. Simultaneously, the molecular interactions of 4-ABPA with evolving crystal facets modulate the growth kinetics, preferentially steering crystallization along the (001) orientation, a conclusion corroborated by the enhanced (001)/(011) diffraction intensity ratio in XRD analysis. Quantitative analysis (Fig. S4) indicates that the average grain diameter (D_{avg}) increases from 344.4 nm for the control to 431.1 nm with the additive, suggesting a reduction in GB density and a lower concentration of bulk defects. Cross-sectional SEM images (Fig. S5) show that the 4-ABPA modification encourages the formation of

vertically aligned, columnar PVK grains that span the entire thickness of the film, contrasting with the irregular morphology of the control. This oriented microstructure reduces grain boundary recombination, thereby enhancing out-of-plane charge transport.²⁴ Water contact angle measurements (Fig. S6) reveal an increase in hydrophobicity, rising from 46.2° for the control to 56.4° for the modified film, which suggests improved moisture resistance.²⁹

Figure 3(a) illustrates the energy-level alignment of PSCs with and without the addition of 4-ABPA in the PVK layer. The pristine PVK shows a valence band maximum (VBM) of -5.94 eV and a conduction band minimum (CBM) of -4.45 eV. After incorporating 4-ABPA, these energy levels shift to -5.87 eV for the VBM and -4.28 eV for the CBM, indicating an enhancement in interfacial energy alignment. This change, which can be attributed to defect passivation and the formation of interfacial dipoles, promotes more efficient charge separation and reduces non-radiative recombination. The hole transport layer, Me-4PACz ($-5.60/-1.60$ eV), aligns well with both NiO_x and the modified PVK layer. Simultaneously, the electron transport materials (PCBM: $-6.00/-3.95$ eV; BCP: $-6.50/-3.50$ eV) facilitate effective electron extraction and hole blocking. The energy-level tuning induced by 4-ABPA enhances charge carrier dynamics, thereby improving photovoltaic performance.^{39,40} To experimentally validate these energy-level shifts, ultraviolet photoelectron spectroscopy (UPS) measurements were conducted (Fig. S7). The secondary electron cutoff (SECO) and valence band edge locations were determined through linear extrapolation of the onset regions in the UPS spectra. For the pristine PVK film, the work function (Φ) was found to be 4.45 eV, and the VBM was 5.94 eV. Following the introduction of 4-ABPA, both parameters exhibited significant shifts: Φ increased to 4.62 eV, and the VBM shifted to 5.87 eV. This corresponds to a reduction in the energy difference ($E_F - E_{\text{VBM}}$) from 1.49 to 1.25 eV,

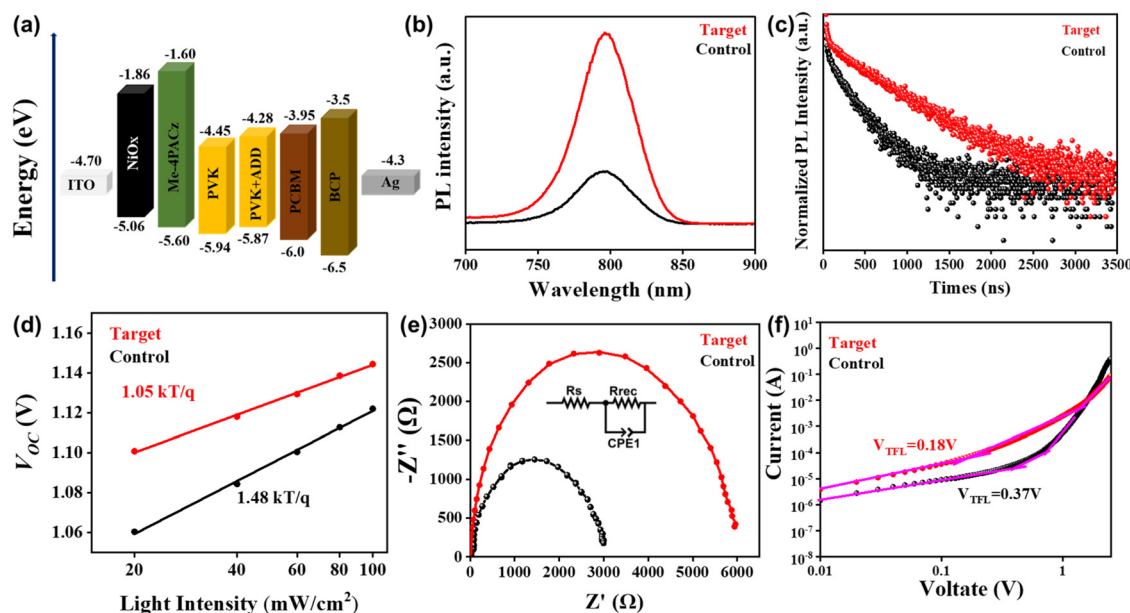


FIG. 3. Energy-level alignment and photophysical characteristics of control and 4-ABPA-modified PVK films and devices. (a) Energy-level diagram of the device structure. (b) Steady-state photoluminescence (PL) spectra. (c) Time-resolved photoluminescence (TRPL) decay curves. (d) V_{oc} as a function of light intensity. (e) Nyquist plots measured at a 0.8 V bias under dark conditions. (f) Dark current density–voltage (J – V) characteristics of electron-only devices.

indicating a shift toward more p-type character. This apparent contradiction given the electron-donating nature of the -NH_2 group is resolved by considering the synergistic action of both functional groups. While the amine moiety can donate electrons, the dominant effect is the phosphonic-acid-mediated passivation of undercoordinated Pb^{2+} defects. The elimination of these deep electron traps significantly reduces the intrinsic n-type background of the PVK. Concurrently, the combined interactions of $\text{-PO}_3\text{H}_2$ and -NH_2 with the PVK lattice establish a surface dipole layer, as evidenced by the work function increase from 4.45 to 4.62 eV. This dipole formation, coupled with defect reduction, collectively shifts E_F toward the valence band, resulting in the observed p-type character while maintaining favorable energy alignment for charge extraction. These findings confirm the formation of surface dipoles and defect passivation effects resulting from 4-ABPA, which collectively enhance interfacial energy alignment and facilitate efficient charge extraction.^{24,41}

Motivated by the improved morphology and crystallization behavior induced by 4-ABPA, we conducted a detailed investigation of the optoelectronic properties of the PVK films. As shown in Fig. S8, the UV-vis absorption spectra reveal a notable enhancement in light absorption for the film incorporating 4-ABPA compared to the control. The passivation effect of 4-ABPA was thoroughly examined through steady-state photoluminescence (PL) and time-resolved photoluminescence (TRPL) measurements. As shown in Fig. 3(b), 4-ABPA incorporation leads to a significant increase in PL intensity, indicating a reduction in defect density and suppressed charge carrier recombination within the PVK films.^{19,42} Furthermore, TRPL analysis in Fig. 3(c) reveals that the 4-ABPA-treated films exhibit prolonged carrier lifetimes compared to the untreated control, consistent with the observed PL quenching behavior. To quantitatively analyze the recombination dynamics, the TRPL decay curves were fitted using a biexponential decay model (supplementary material).^{43–45} Notably, the target films exhibit an extended average lifetime of 506.947 ns in comparison to the control film, which is 119.139 ns. These findings suggest that non-radiative recombination is considerably reduced in films with 4-ABPA, attributed to the superior passivation effect of 4-ABPA and the improved quality of the modified film. The fitting parameters are summarized in Table S2 in the supplementary material. Light intensity-dependent measurements of V_{OC} [Fig. 3(d)] further reveal suppressed Shockley-Read-Hall recombination.⁴⁶ The slope of V_{OC} vs $\log P$ decreases from $1.48 k_B T/q$ to $1.05 k_B T/q$, indicating reduced Shockley-Read-Hall recombination.⁴⁷ Additionally, the power-law exponent α in $J_{SC} \propto P^\alpha$ (Fig. S9 in the supplementary material) increases from 0.97 to 0.98, suggesting lower bimolecular recombination.⁴⁸ Electrochemical impedance spectroscopy (EIS) measurements under dark conditions [Fig. 3(e)] reveal two semicircles corresponding to series resistance (R_s) and recombination resistance (R_{rec}). Incorporation of 4-ABPA reduces R_s from 21.7Ω to 18.2Ω , indicating improved interfacial charge transport, while increasing R_{rec} from 2973.8Ω to 5920.9Ω , confirming suppressed non-radiative recombination. The equivalent circuit model used for fitting is shown in Fig. 3(e), with the extracted parameters summarized in Table S3.^{49,50} To evaluate the defect passivation effect of 4-ABPA, dark current-voltage (J - V) characteristics of electron-only devices were measured [Fig. 3(f)], with configuration of ITO/ SnO_2 /PVK/PCBM/Ag.^{41,51} The J - V curves exhibit three distinct regimes—Ohmic, trap-filled limit (TFL), and Child's law—with the TFL voltage V_{TFL} decreasing from

0.37 V (control) to 0.18 V (4-ABPA).^{52,53} This reduction in V_{TFL} confirms a substantial decrease in trap density after 4-ABPA treatment. These results demonstrate that 4-ABPA effectively passivates defects and suppresses recombination, thereby enhancing device performance.

To evaluate the influence of 4-ABPA incorporation on the photovoltaic performance of PSCs, devices were fabricated with the configuration ITO/Me-2PACz/PVK/PEABr/PC₆₁BM/BCP/Ag, as illustrated in Fig. S10. The optimal concentration of 4-ABPA was determined to be 1 mg/mL following a series of optimizations within the range of 0–1.5 mg/mL, with the results summarized in Table S3 and Fig. S11. As shown in the J - V characteristics in Fig. 4(a), the 4-ABPA-modified device achieved a champion PCE of 24.1% ($V_{OC} = 1.15$ V, $J_{SC} = 25.03$ mA/cm², FF = 83.73%), compared to 22.16% ($V_{OC} = 1.12$ V, $J_{SC} = 24.35$ mA/cm², FF = 81.26%) for the control device. Figure S11 confirms the improvement in PCE of the 4-ABPA-treated devices, with the highest device performance reaching 24.1%, along with enhancements in J_{SC} , V_{OC} , and FF. The device performance showed excellent reproducibility, with 24 independently fabricated devices at the optimal 4-ABPA concentration (1 mg/mL) yielding an average PCE of $23.72 \pm 0.38\%$. To assess device reliability and the accuracy of the PCE measurement, we performed hysteresis analysis and stabilized power output (SPO) measurements. As shown in Fig. S12, the J - V curves under forward and reverse scanning reveal a significantly reduced hysteresis in the 4-ABPA-modified device compared to the control. The calculated hysteresis index (HI) decreased substantially from 2.4% for the control device to only 0.9% for the 4-ABPA-treated device. This marked reduction indicates diminished ionic migration and trap-assisted recombination, leading to more efficient charge extraction. Furthermore, the stabilized power output measured at the maximum power point confirmed excellent operational stability. The modified device maintained a high steady-state PCE of 23.89% over a standard tracking period of 500 s, outperforming the control device, which stabilized at 21.78% under the same conditions (Fig. S13). These results collectively validate the champion PCE of 24.1% obtained from the J - V scan. This improvement is attributed to the ability of 4-ABPA to passivate defects and modulate interfacial dipoles. This modification reduces charge recombination losses and facilitates charge extraction and transport, resulting in enhanced charge separation and collection, ultimately leading to a higher PCE. The external quantum efficiency (EQE) spectra in Fig. 4(b) corroborate the J_{SC} values, revealing integrated current densities of 24.2 mA/cm² for the 4-ABPA-modified device and 25.01 mA/cm² for the control device, consistent with the J - V data.

To evaluate the stability of 4-ABPA-modified PSCs, we investigated their long-term stability under three stress conditions: ambient dark storage at 10 °C with approximately 20% relative humidity (RH), thermal stress at 65 °C in nitrogen (N_2), and continuous illumination at AM 1.5 G with an intensity of 100 mW/cm² in nitrogen.^{29,41,50} Unencapsulated 4-ABPA devices retained 92% of their initial PCE after 1200 h, compared to 85% for control devices [Fig. 4(c)]. Under thermal aging at 65 °C, the 4-ABPA devices maintained 87% PCE after 600 h, while the control devices exhibited 75% [Fig. 4(d)]. In the continuous illumination test, the 4-ABPA devices preserved 85% efficiency after 840 h, whereas that for the control devices dropped below 85% by 400 h [Fig. 4(e)]. This improvement is attributed to the phosphonic acid group in 4-ABPA, which passivates undercoordinated Pb^{2+} at the

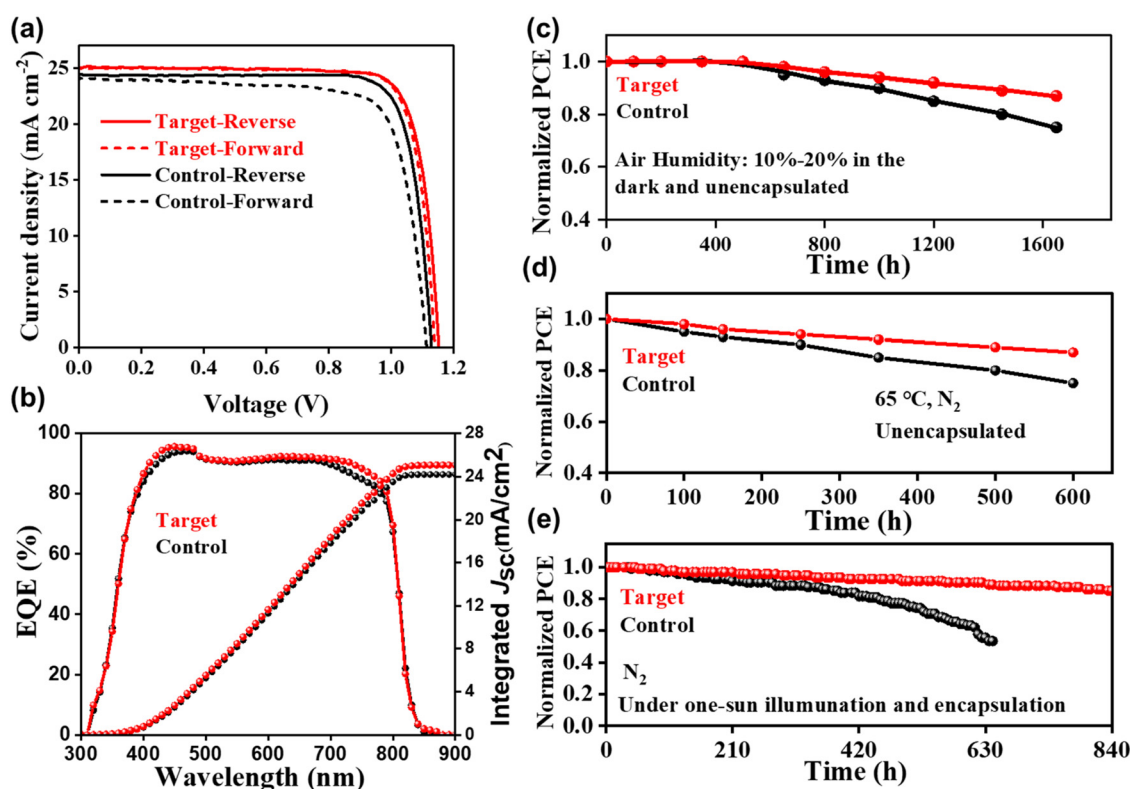


FIG. 4. Photovoltaic performance, recombination analysis, and stability assessment of control and 4-ABPA-modified devices. (a) J–V curves of the best-performing devices. (b) External quantum efficiency (EQE) spectra with integrated photocurrent. (c) Dark storage stability in ambient air without encapsulation. (d) Thermal stability at 65 °C in N₂. (e) Light-soaking stability under 100 mW/cm² illumination in N₂.

interfaces, effectively suppressing non-radiative recombination and ion migration. -NH₂ also works with the other groups to relieve lattice strain, which improves charge dynamics.²⁴ The synergistic defect passivation and stress alleviation provided by 4-ABPA's dual-functional groups yield performance metrics that compare favorably with champion inverted PSCs reported recently, while offering the distinct advantage of addressing both electronic and mechanical degradation pathways in a single molecular additive.⁵⁴ This dual functionality enables scalable fabrication of stable, high-performance PVK modules. Importantly, our work distinguishes 4-ABPA from its salt form, 4-ABPACl, which was previously used as a grain boundary crosslinker in mesoscopic PVK devices.⁵⁵ Unlike the surface-limited hydrogen bonding of 4-ABPACl, 4-ABPA engages in bulk Lewis acid–base coordination with Pb²⁺ sites, enabling simultaneous defect passivation, stress relief, and crystallization control key advances for inverted PSCs not addressed in prior surface-engineering approaches.

To summarize, our results show that 4-ABPA is a multifunctional additive for inverted PSCs because it tackles two key issues at once: healing defects and guiding better crystal formation. The -PO₃H₂ and -NH₂ groups work together to passivate undercoordinated Pb²⁺ and I⁻ vacancies, while the flexible alkyl chain alleviates lattice strain. GIXRD measurements show that this molecular structure lowers residual stress by 64%, and TRPL measurements show that it increases the lifetime of charge carriers from 119 ns to 507 ns. The optimized devices

achieved a champion PCE of 24.1% (V_{OC} = 1.15 V, J_{SC} = 25.03 mA/cm², FF = 83.73%) and demonstrated excellent operational stability. After 840 h of continuous 1-sun illumination at 65 °C, they still retained 85% of their initial performance. This study proposes a simple yet effective molecular approach for the advancement of stable, high-efficiency PVK photovoltaics.

See the [supplementary material](#) for more details on film preparation and device fabrication, as well as FTIR, XRD, SEM, UPS, UV-vis, J–V, stabilized power output, contact angle measurements, and statistical analysis of device performance.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Ibrahim Albrahe: Data curation (equal); Investigation (equal); Methodology (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Haibib Aziguli**: Data curation (equal); Formal analysis (equal); Investigation (equal); Resources (equal); Validation (equal); Writing – review & editing (equal). **Tengfei Zhang**: Data curation (supporting); Formal analysis (supporting); Methodology (supporting); Project administration (equal); Validation (supporting); Writing – original draft (equal); Writing – review & editing (equal). **Liwei Chen**: Data curation (supporting); Formal analysis (supporting); Investigation (supporting); Methodology (supporting); Validation (supporting); Writing – original draft (supporting). **Hailong Li**: Data curation (supporting); Formal analysis (supporting); Investigation (supporting); Methodology (supporting); Validation (supporting); Writing – original draft (supporting); Writing – review & editing (supporting). **Yue Yu**: Data curation (supporting); Formal analysis (supporting); Investigation (supporting); Methodology (supporting); Validation (supporting); Writing – review & editing (supporting). **Jiangzhao Chen**: Data curation (equal); Funding acquisition (equal); Investigation (equal); Project administration (equal); Validation (equal); Writing – review & editing (equal). **Juan Hou**: Formal analysis (equal); Funding acquisition (equal); Project administration (equal); Supervision (lead); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available within the article and its [supplementary material](#).

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